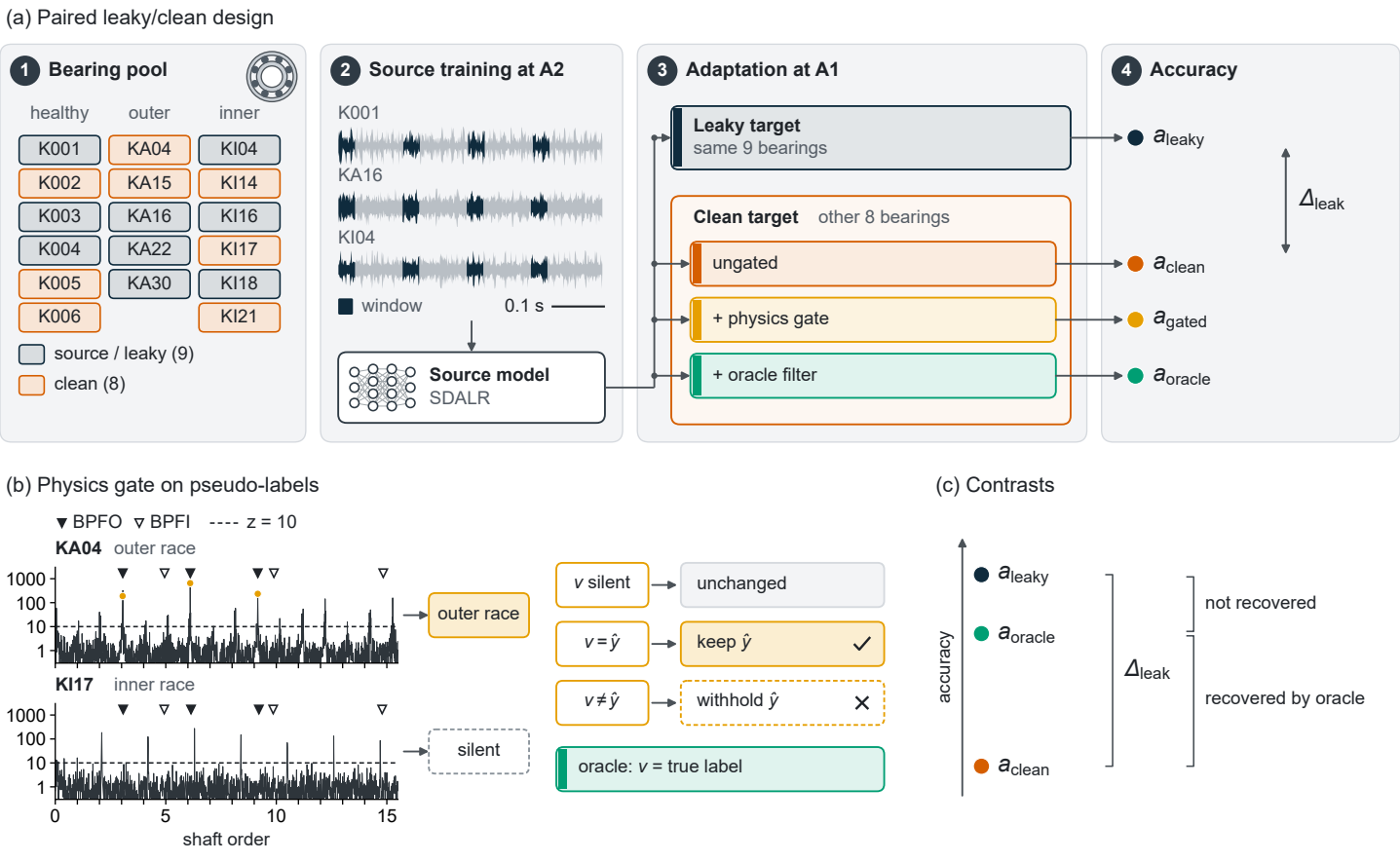

(a) Paired leaky/clean design
1 Bearing pool
healthy
outer
inner
K001
K002
K003
K004
K005
K006
KA04
KA15
KA16
KA22
KA30
KI04
KI14
KI16
KI17
KI18
KI21
source / leaky (9)
clean (8)
2 Source training at A2
K001
KA16
KI04
window
0.1 s
Source model
SDALR
3 Adaptation at A1
Leaky target
same 9 bearings
Clean target
other 8 bearings
ungated
+ physics gate
+ oracle filter
4 Accuracy
$a_{leaky}$
$a_{clean}$
$a_{gated}$
$a_{oracle}$
$\Delta_{leak}$
(b) Physics gate on pseudo-labels
▼ BPFO ▽ BPFI ---- z = 10
KA04 outer race
KI17 inner race
1000
100
10
1
0
5
10
15
shaft order
outer race
silent
v silent
unchanged
v = ŷ
keep ŷ
v ≠ ŷ
withhold ŷ
oracle: v = true label
(c) Contrasts
accuracy
$a_{leaky}$
$a_{oracle}$
$a_{clean}$
$\Delta_{leak}$
not recovered
recovered by oracle